
EE/CprE/SE 492 WEEKLY REPORT 5

October 16, 2025 - October 30, 2025

Group number: 16

Project title: Outdoor Roomba

Client &/Advisor: Prof. Diane Rover

Team Members/Role:

Brodie M Bates - **Embedded system**

Daniel Vergara Pinilla - **Machine Learning**

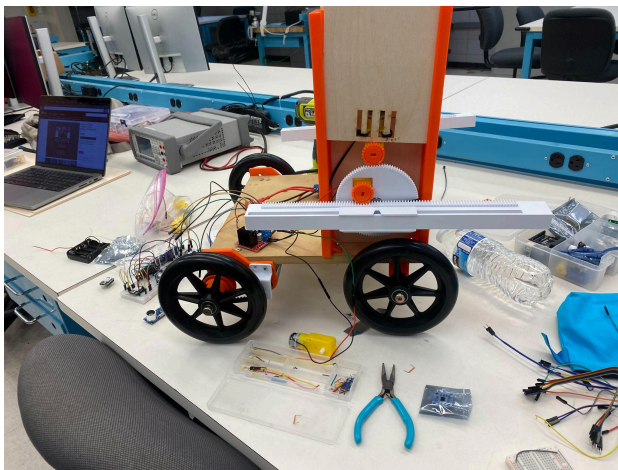
Mitchell D Owen - **Project Tester**

Om Shukla - **Frontend Developer**

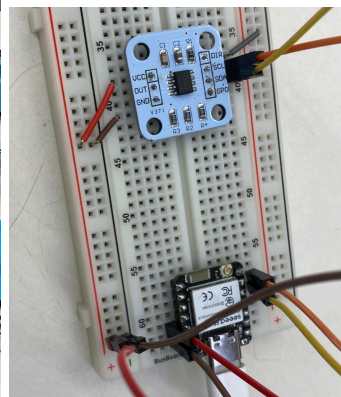
Sudipta Halder - **Simulation developer**

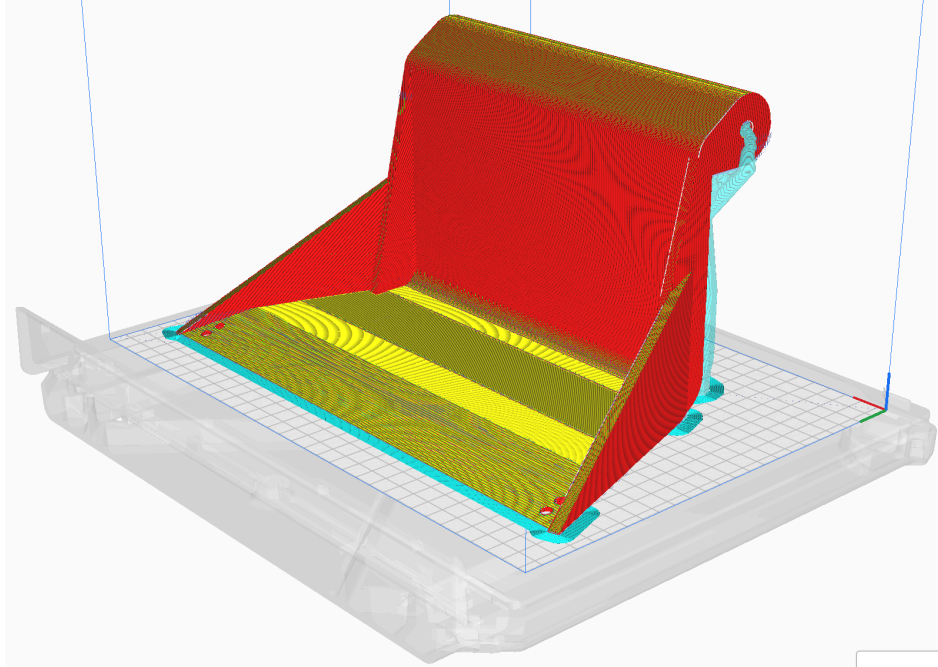
○ Weekly Summary

This week we created the arms for our power shovel mechanism. We had to reprint the arm holder 5 times due to minor errors that need to be fixed at every print. Then we work on redesigning it for it to be ready for printing. We started working on a navigation milestone for this project. This is the final milestone for our robot that needs to be completed. This milestone will require implementation of the server and all other microcontrollers code to make navigation and logic of our roomba work. If the shovel works as designed before week 11 we should have a chance to complete the project on week 13. For user testing we are planning to have it be used by the Halder family executing sudipta halder for their thanksgiving event.



Arm and encoder





Printing of the shovel

○ **Past weeks accomplishments**

- Sudipta Halder: Designed and printed the arms and had them work successfully. Redesigned a shovel to be printed strong enough to lift cans and bottles and have we wide enough to pick it in. Designed and printed holder for magnets used by the magnetic encoders for the motors.
- Brodie Bates: Wrote code to make the arm extend, retract, and move up and down. Wrote code to take in sensor data for four ping sensors to be used on the bottom of the robot. Started working on I2C for our motor controllers. Tested parts of the arm with the code.
- Daniel Pinilla: Made sure the database and websockets worked well for the group by fixing errors. Have the website up on their server at ETG and started working on adding the functionality for static and trash objects for the server map .
- Om Shukla: Finished login functionality and uploaded it to the server. Also finished the redesign of the frontend to look more professional.
- Mitchell Owen: assembling the arm mechanic and making improvements in the chassis to make it wider and more effective and adjusting bracket sizing. Designing the power shower mechanics.

- **Pending issues**

- Om Shukla: Currently waiting for the server to update with the new frontend code so we can test the login functionality.
- Sudipta Halder: work outside of senior design will get in the way for this project. Many have exams in the next two weeks. It is hard to have everyone work together. Hopefully implementation goes well else it can interfere with other's jobs and classes.

○ **Individual contributions**

<u>NAME</u>	<u>Individual Contributions</u>	<u>Hours this week</u>	<u>HOURS cumulative</u>
Brodie Bates	Wrote code for the arm, ping sensors, and I2C motor encoders. Tested each.	18	57
Om Shukla	Finished login functionality and website redesign.	15	36
Sudipta Halder	Printing and testing multiple iterations of the power shovel arm's mechanism. Multiple Simulation of the gears of the power shovel and the shovel it itself.	40	83
Daniel Mauricio Vergara	Worked on database and websockets	8	38
Mitchell Owen	Assembling the arm mechanic and adjusting chassis/Electronics holder.	15	45

○ **Plans for the upcoming week**

- Sudipta Halder: implement the shovel, start Milistone 4: Navigation. Program the raspberry pi to work with server and the current implementation of the machine learning model/ and make sure that data front the sensor and decisions for the raspberry pi meets deadlines.
- Brodie Bates: Continue to work on the I2C encoders, and connect the second arm. I'll work on making all devices needed to drive the robot into a permanent place.
- Om Shukla: Start working on endpoints to retrieve roomba information from the server.
- Mitchell Owen: Having the power showed attached and properly working.
- Daniel Pinilla: Start working on implementation with the robot and the server communication. Have the machine learning code work in union with other programs .

- **Summary of weekly advisor meeting**

The first week of this report we had no advisor meeting. During the second advisor meeting we talked about how we need to use plexiglass to demo so that the internals are visible. Then we talked about how we know the cables will work for our design to lift the power shovel up? The cable is thick to wind, might need a specific cable to be better at winding. Will it have different shapes when it gets loose? Check for other systems that use a winding cable to pick things up. Look into buying a part that would do this as opposed to 3d printing everything. The smaller the cylinder to wind around is better. Think through the winding cable idea to make sure it will work. Possible solution is use motors for shovel or be able to somehow pull / push it where it needs to go. Create a solution simultaneously as the cable one with a motor direct drive. Don't have much room for actual cans to go in, cut orange things in half, extend back tires out further to avoid cans getting stuck underneath the robot.

Then we talked about SLAM - simultaneous location and mapping. Used in autonomous robots, self driving cars, and others. May have disadvantages. Do not have to implement just have the mapping part done so it will accept a map that we give it. Will log location of objects.

Next meeting we need to show the diagram of behavior between low level code and high level code, such as messages back and forth between server and pi and low level micro controllers. Software Behavior diagram for the detection software using various sensors. How those two communicate. How we are synchronizing and communicating between high and low levels. Formally identify tasks and threads of the robot. Systems have tasks running between systems, we need to think how this will listen to the other tasks and how they interact with each other.

Make it clear to the industry panel about the functionality of robots. Tasks are working over time and they will synchronize eventually, raise flags or communicate.